MILITARY SPECIFICATION

CUSHION, CRUTCH

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers three types of support cushions, two underarm and one thorax, for use with wooden crutch armpieces (see 6.1).

1.2 Classification. The crutch design shall be one of the following types:

Type I - Cushion with fabric reinforcement, underarm.
Type II - Cushion without fabric reinforcement, underarm.
Type III - Cushion contoured for rib cage conformity.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-676 - Boxes, Setup.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6515

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

DRAWING

Defense Personnel Support Center (DPSC) Drawing

No. 1792 CUSHION, CRUTCH, AXILLARY
No. 22210 CUSHION, CRUTCH

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Rubber or plastic compound. The cushion shall be composed of natural, reclaimed, or synthetic rubber, suitable in all respects for obtaining a good-quality sponge. As an alternate, for Types I and II only, a suitable plastic cushioning material shall be used. The molded compound (sponge rubber or plastic) shall be free from bleeding, blooming, tackiness or objectionable odor and shall not contain known skin irritants.

3.1.2 Fabric. The fabric used shall be high strength cotton or linen cloth suitable for use as reinforcement.

3.2 Design and dimensions. Design and dimenisons of the cushions shall be as shown on DPSC Drawings Nos. 1792 and 22210.

3.2.1 Structure. The cushions shall be molded of sponge rubber having interconnecting cells or a suitable closed cell plastic. The Type I shall be molded around a strong fabric reinforcement. The fabric shall not show through the sponge at any point. Cells shall be reasonably uniform in size, of a firm character and free from large voids. The cushion surface shall be a thin porous skin or rind.

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3.3 Physical requirements.

3.3.1 Accelerated aging. When aged as described in 4.4.2, the cushion shall not become sticky or show blistering or bleeding, and shall not crack when bent sharply back upon itself.

3.3.2 Weight. The weight of the cushions shall be as follows:

Type I	-	95 grams, minimum.
Type II	-	100 grams, minimum.
Type III	-	200 grams, minimum.

3.3.3 Color. Color of the cushion shall be tan or light brown.

3.4 Workmanship. The workmanship shall be first class throughout. The crutch cushions shall be free from defects which detract from their appearance or impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

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4.1.3 Certificates of quality. Certificates of quality supplied by the manufacturer of the rubber, plastic, or fabric may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and as indicated in table I. Unit of product for examination purposes shall be one cushion.

TABLE I. Sampling criteria for examination.

Type of defect	Inspection level	AQL (percent defective)
Major	II	1.0
Minor	II	4.0
Dimensional	S-2	2.5

4.2.2 For tests. Sampling for tests shall be in accordance with MIL-STD-105 and as indicated in table II. Unit of product for test purposes shall be one cushion. Where feasible, the same sample unit shall be used for the determination of two or more characteristics.

TABLE II. Sampling criteria for tests.

Characteristic	Requirement par.	Test procedure	Inspection level	AQL (percent defective
Accelerated aging	3.3.1	4.4.2	S-1	1.0
Weight	3.3.2	4.4.1	S-1	1.0

4.3 Examination of cushion. The cushion sample selected in accordance with 4.2.1 shall be examined for compliance with specification requirements. The examination shall be conducted in accordance with but not restricted to defects listed in table III. Additional defects shall be classified in accordance with MIL-STD-105.

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TABLE III. Classification of defects.

Category	Defects
Major:	
101	Cushions do not fit over or adequately grip the specified crutch armpiece.
102	Cushions not free of large voids.
103	Absence of thin porous skin or rind.
104	Tackiness apparent upon touch.
105	Cushions are other than specified design.
Minor:	
201	Fabric shows through (Type I).
202	Color not uniform, or not as specified.
203	Objectionable odor.
204	Cushion not of a firm character.

4.3.1 Dimensional examination. Any dimension not within the specified tolerance shall be classified as a defect.

4.3.2 Packaging inspection. The inspection of the unit, intermediate package, packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.4 Tests. Samples selected in accordance with 4.2.2 shall be tested to determine compliance with specification requirements.

4.4.1 Weight. Cushions shall be weighed on a suitable scale or balance, accurate to the first decimal place, to determine compliance with minimum weight requirements.

4.4.2 Accelerated aging. Each sample shall be aged in an oxygen bomb for 96 hours at $158^{0} \pm 2^{0}$ F ($70^{0} \pm 1^{0}$ C) and a pressure of 300 $\pm$ 10 pounds per square inch. After aging, the bomb shall be removed from the heating chamber, cooled, and oxygen released very slowly. The sample shall be removed and examined for tackiness, blistering, bleeding and other deleterious effects. The sample shall be placed in a dessicator for at least 16 hours, removed and bent sharply back upon itself. The material shall not crack.

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5. PACKAGING.

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Unit of issue. One (1) pair of cushions, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Unit package. Each pair of cushions of the type specified shall be individually wrapped in tissue paper and packaged in a box constructed in accordance with PPP-B-566 or PPP-B-676. Closure shall be as specified in the appendix of the applicable box specification.

5.1.2.2 Intermediate package. Six unit packages shall be packaged in a box conforming to PPP-B-566, PPP-B-676, or PPP-B-636, type CF, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.1.3 Level C. Crutch cushions shall be clean and dry. One pair of cushions shall be packaged in a manner that will afford adequate protection against deterioration and physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Twelve intermediate packages (72 units) shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class weather-resistant. Closure, strapping and waterproofing shall be as specified in the appendix of the box specification.

5.2.2 Level B. Twelve intermediate packages shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

5.2.3 Level C. The packaged cushions shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carriers rules and regulations.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. Contract or purchase order number and name of the contractor shall be shown.

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5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Types I and II cushions are intended for use on the wooden crutch armpiece so that a resilient padding is provided for the body weight. Type III cushion is designed to be used with the wooden crutch against the rib cage, and not directly under the armpit.

6.2 Ordering data. Ordering data should include the following:

a. Title, number and date of this specification.
b. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
c. National Stock Number.

6.3 National stock number coverage. This specification covers the following item appearing in the Federal Supply Catalog:

National Stock Number	Item Identification
6515-00-777-7340	CUSHION, CRUTCH, Axillary
6515-01-042-0427	CUSHION, CRUTCH, Thoracic

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
DLA-DM

Project Number 6515-2031

Review activity:
DOD - MB

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE
MIL-C-37578 CUSHION, CRUTCH, AXILLARY

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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